

Lunchtime lecture series - #39

8 October 2024

Technology Challenges to Meet Sustainability Goals of Concrete Used
for Segmental Tunnel Linings

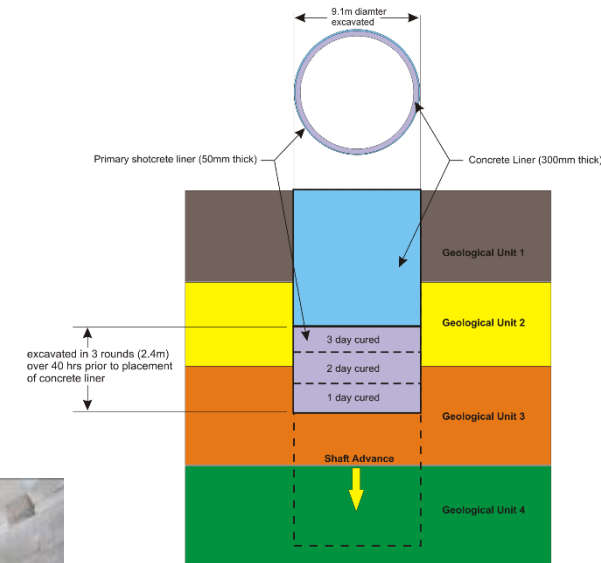
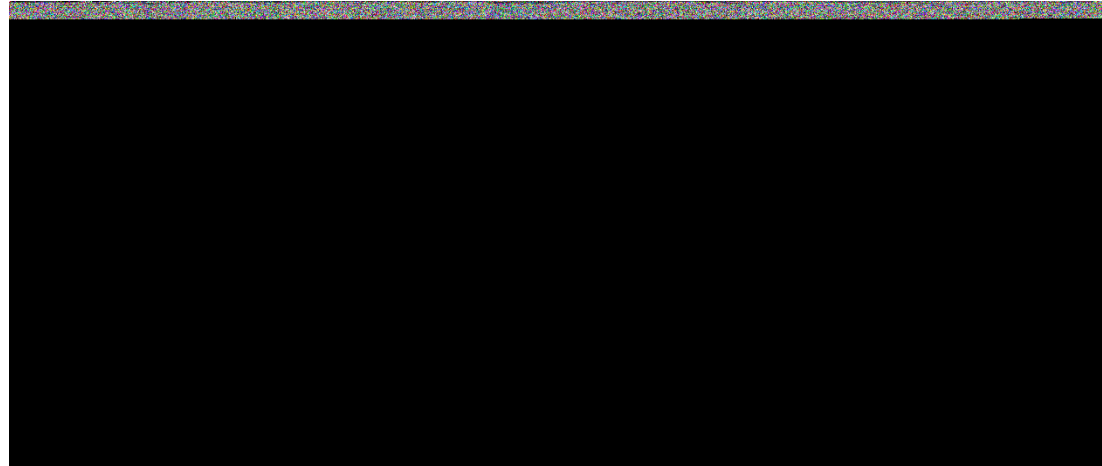
Barzin Mobasher, PhD, PE, FACI, F-RILEM

**Professor, School of Sustainable Engineering and the Built Environment
Arizona State University, Tempe, AZ 85287-5306**

Entry into the underground aspects of concrete application

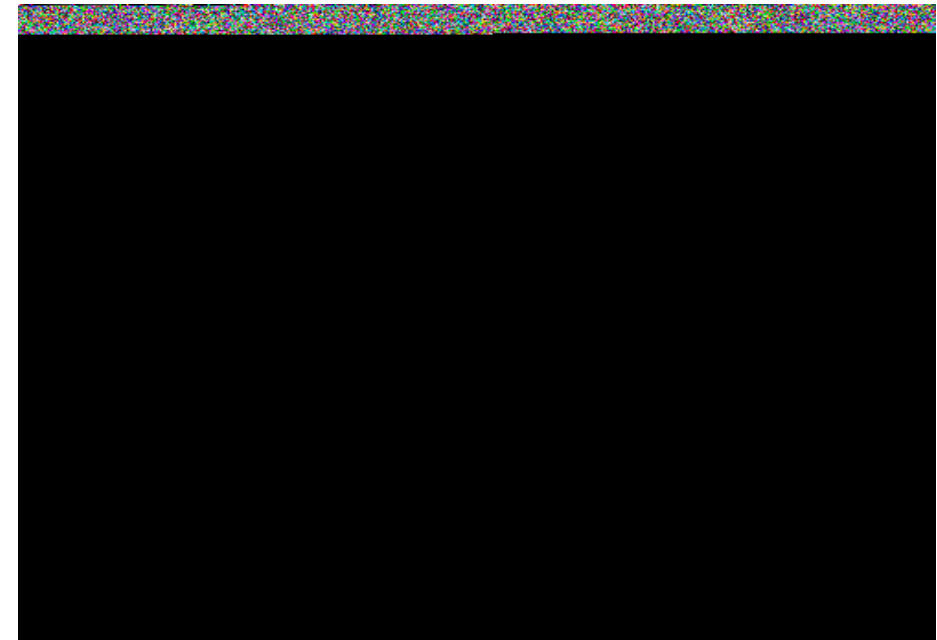
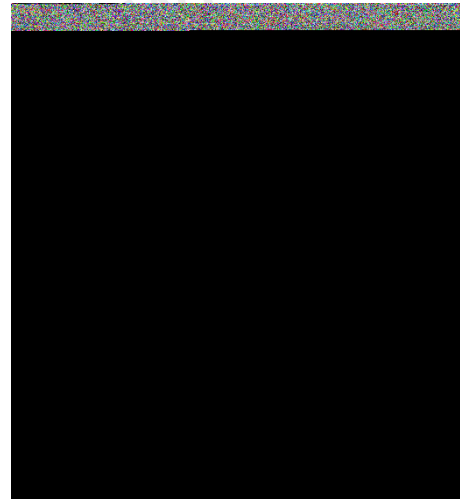
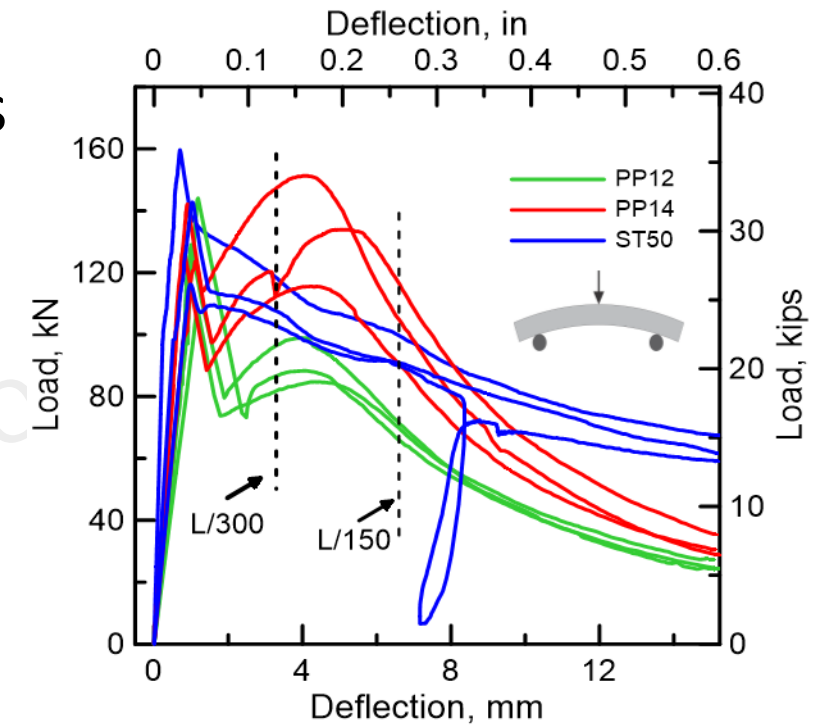
Resolution Copper – Magma Copper mine, Superior, Arizona

- Evaluation of the design of initial fiber-reinforced shotcrete support for a 2000 m-deep shaft.



Focus on FRC Tunnel Lining Engineering Applications

- Ductility, Durability, Sustainability
- Increased energy absorption, fatigue life, seismic, impact conditions



The Hypothesis

#1: Changing Societal Attitude towards Concrete

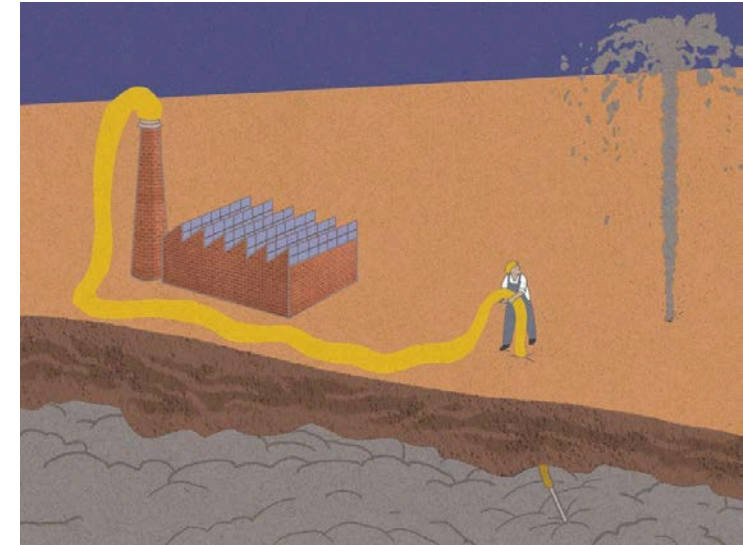
#2: Reinforcement Strategy is the Cornerstone of Concrete Sustainability

- China's Cement use in 3 years > US use in the past 100 years [6.6 GT (2011-2013) vs. 4.5 GT (1901-2000)]
- About 60-75% of the concrete used under flexure doesn't carry any load since its contribution is ignored due to cracking
- Very little has been done about efficient structural design of concrete



Guardian, Best of 2019,
Concrete: the most destructive material on Earth

<https://www.theguardian.com/cities/2019/feb/25/concrete-the-most-destructive-material-on-earth>



Scientific American, March 1, 2024,

The False Promise of Carbon Capture as a Climate Solution

Fossil-fuel companies use captured carbon dioxide to extract more fossil fuels, leading to a net increase in atmospheric CO₂

<https://www.scientificamerican.com/article/the-false-promise-of-carbon-capture-as-a-climate-solution/>

High performance Concrete Materials

- Need to Address:

- HPC design principles
- HPC constituents
- Mix design issues
- Production and Curing
- Field issues

- Role of Pozzolons:

- Enhancing Mechanical Properties
- Improving Durability
- Enhancing Constructability
- Producing High-Performance Concrete

- Curing, One pass finishing

- Screed
- Float
- Texture

Fundamentals of Concrete Technology

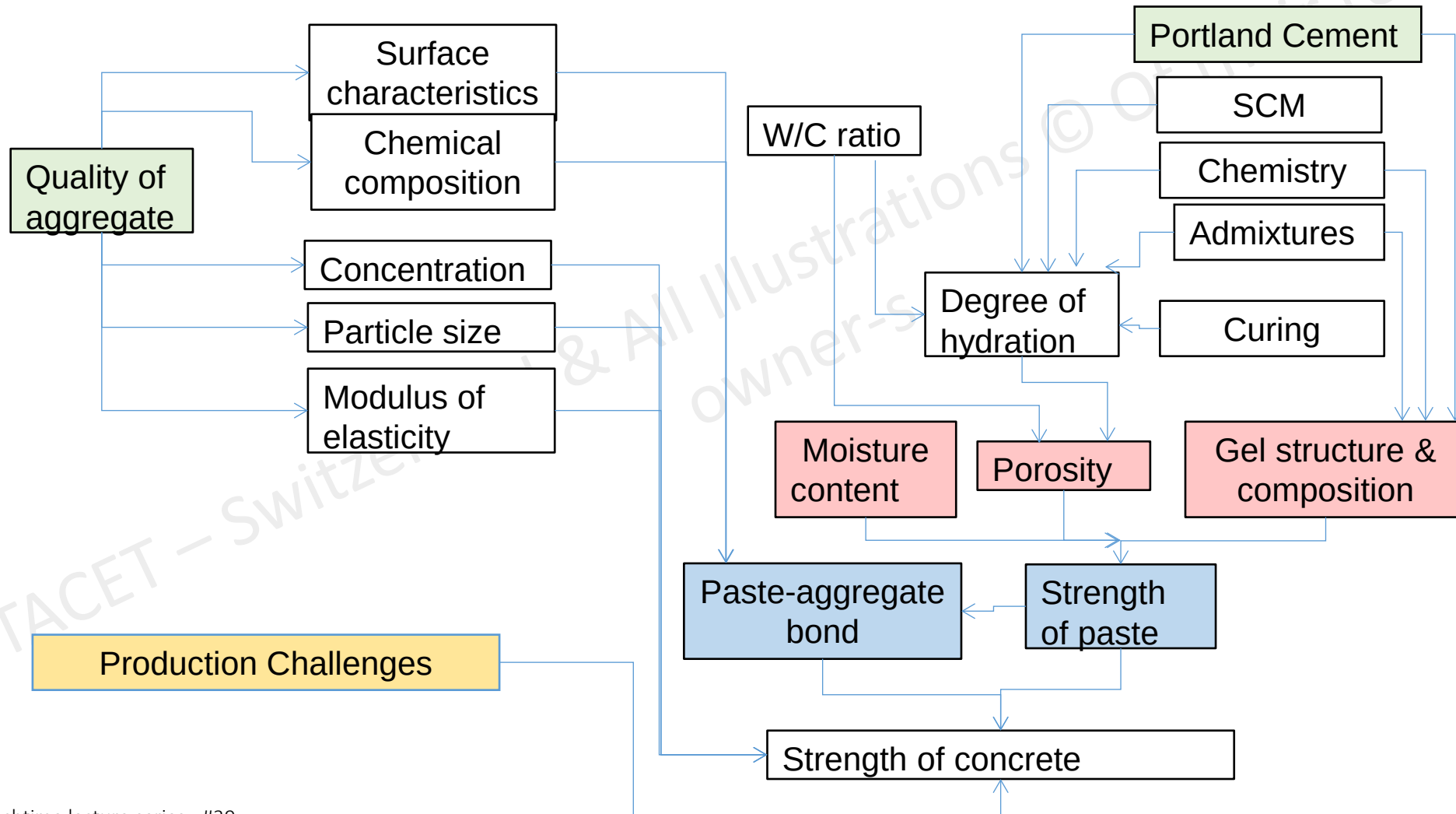
Fresh Concrete

- Constituents: Portland cement, aggregates, water, admixtures
- Mix Design
- Rheology & flow, consolidation, packing factors, slump, bleeding
- Production speed, placing, finishing.
- Curing, early age plastic shrinkage cracking
- Temperature effects

Hardened Concrete

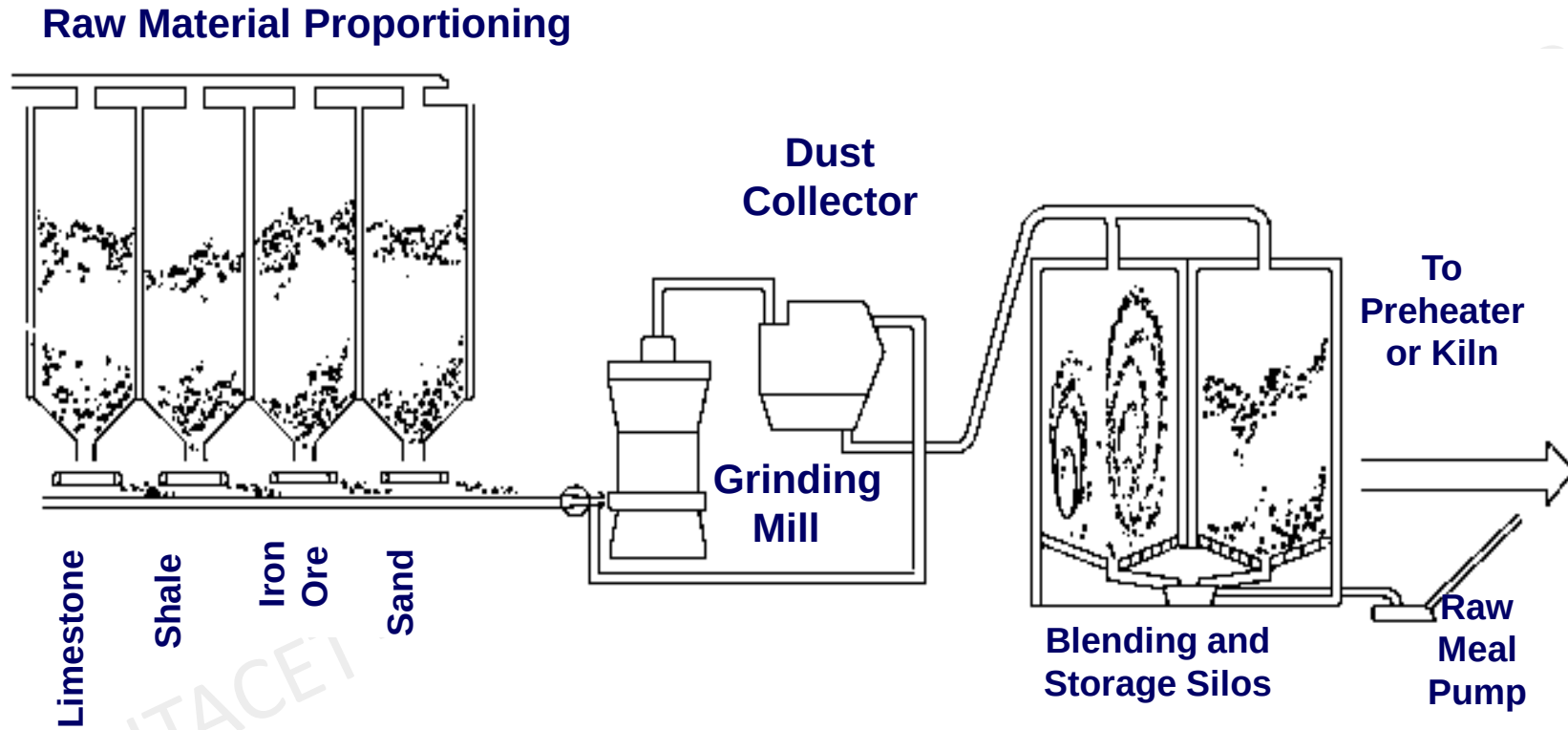
- Factors affecting strength
- Choice of reinforcement
- Physical and chemical properties
- Mix design: Strength, permeability
- Durability, freeze-thaw, creep, shrinkage, permeability, corrosion, alkali-silica reaction, sulfate attack, carbonation, chloride-induced rebar corrosion.
- Specifications and quality control

Factors Affecting Strength of Concrete



Cement manufacturing

Raw Material Proportioning & Blending



After analysis, the raw materials are proportioned, ground to fine powder, and blended. Some cement plants add water to the material during grinding, then blend and store it as a slurry.

Portland Cement Constituents

- Four major constituents plus gypsum:
- $3\text{CaO} \cdot \text{SiO}_2$, C_3S , tricalcium silicate
- $2\text{CaO} \cdot \text{SiO}_2$, C_2S dicalcium silicate
- $3\text{CaO} \cdot \text{Al}_2\text{O}_3$, C_3A tricalcium aluminate
- $4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{Fe}_2\text{O}_3$, C_4AF tetracalcium aluminoferrite
- $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$, $\text{C}\bar{\text{S}}$ Gypsum, Hemi-Hydrate

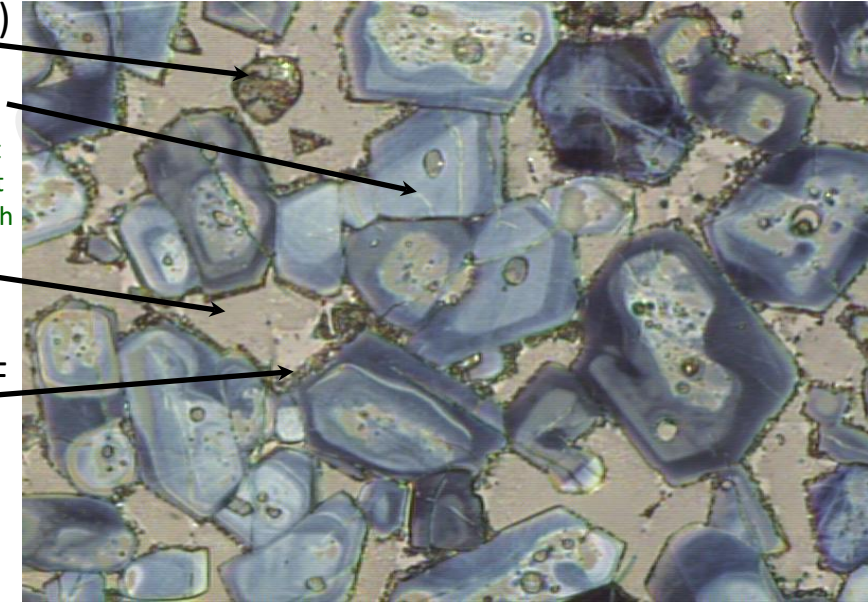
C_2S (belite)

C_3S (alite)

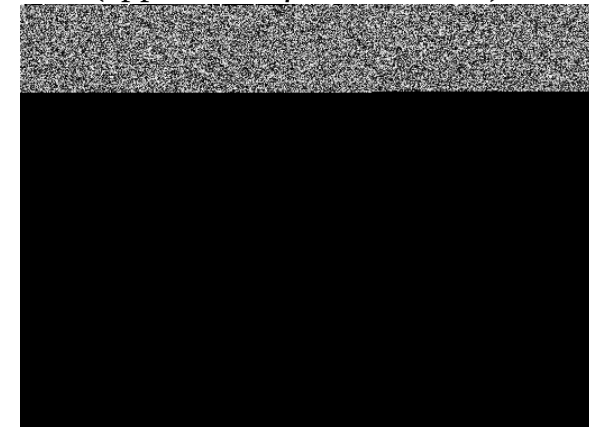
Most important
constituent that
governs strength

C_3A

C_4AF



[BEI by Paul Stutzman, NIST]
(approx 200 μm full width)

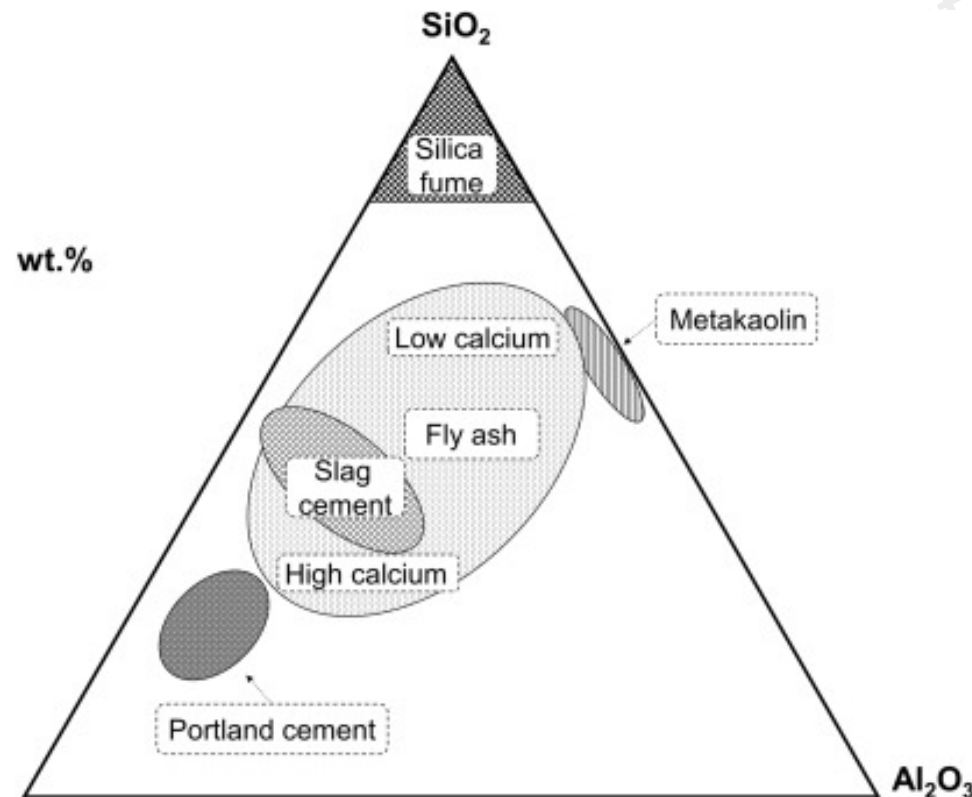


		Compounds, Values in %			
Type	Fineness, m^2/Kg	C_3S	C_2S	C_3A	C_4AF
I	369	54	18	10	8
II	377	55	19	6	11
III	548	55	17	9	8
IV	340	42	32	4	15
V	373	54	22	4	13
White		33	46	15	2

Calcium oxide	64%	CaO	"Lime"	C
Silicon oxide	22%	SiO ₂	"Silica"	S
Aluminum oxide	6%	Al ₂ O ₃	"Alumina"	A
Iron oxide	3%	Fe ₂ O ₃	"Iron"	F

Cement and SCM Domain

- Flyash, Slag, Silica Fume, Metakaolin, limestone, Calcined clay, Harvested Flyash, PLC, or LC3 Cements
- Typical LC3-50 cement blend mixture has 30% calcined clay, 15% limestone, 5% gypsum, and 50% clinker, resulting in 40% carbon emissions reduction and 25% reduction in overall cost

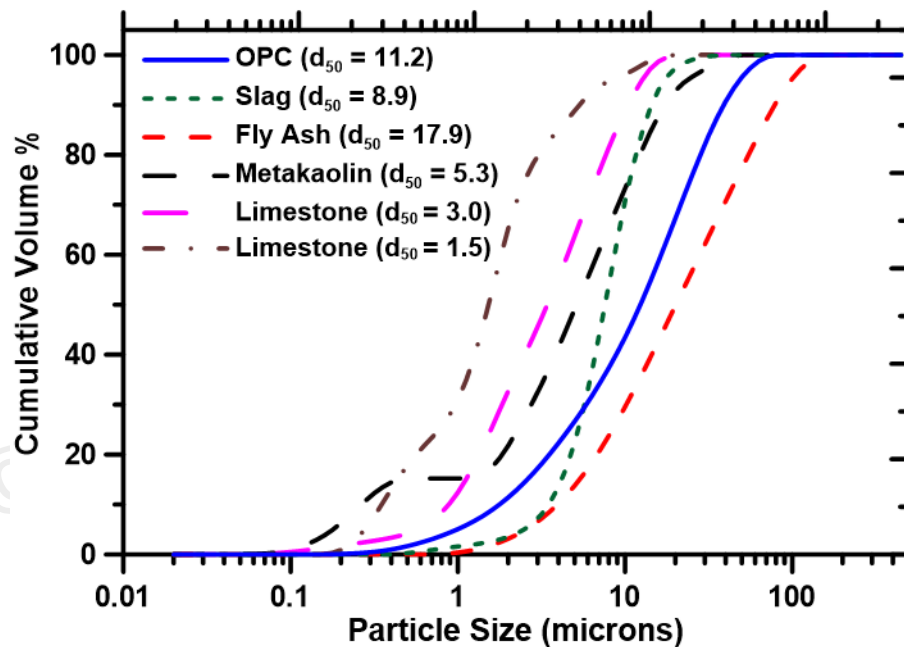


From left to right:

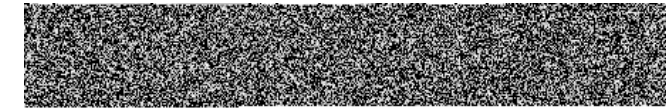
- Fly ash (Class C)
- Metakaolin (calcined clay)
- Silica fume
- Fly ash (Class F)
- Slag Cement
- Calcined shale

Supplementary cementitious materials (SCM)

- Use of Supplementary Cementitious Materials (SCM) is recognized as one of the most important sustainability aspects of concrete industry's effort to address its carbon footprint. (Damtoft et al. 2008a)
- Category of inorganic powders with similar and compatible chemical and mineral characteristics to conventional portland cement.
- ASTM designations: Type I/II ordinary portland cement (OPC) conforming to ASTM C 150, Class F, and C fly ash and metakaolin, or natural pozzolans conforming to ASTM C618, slag conforming to ASTM C 989, limestone powder conforming to ASTM C 568, and microsilica (silica fume) conforming to ASTM C 1240.



Portland Cement Particles

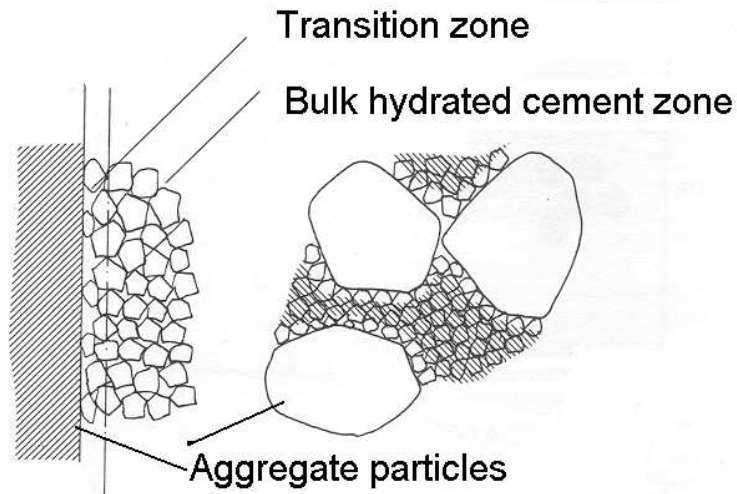


Silica Fume Particles

Physical and Chemical Effects of SCMs on the Interface Transition zone

- Why high durability matters

- *Reduced maintenance costs*
- *Longer life, Serviceability and sustainability*
- *Reduce the rate at which chlorides can enter the concrete.*



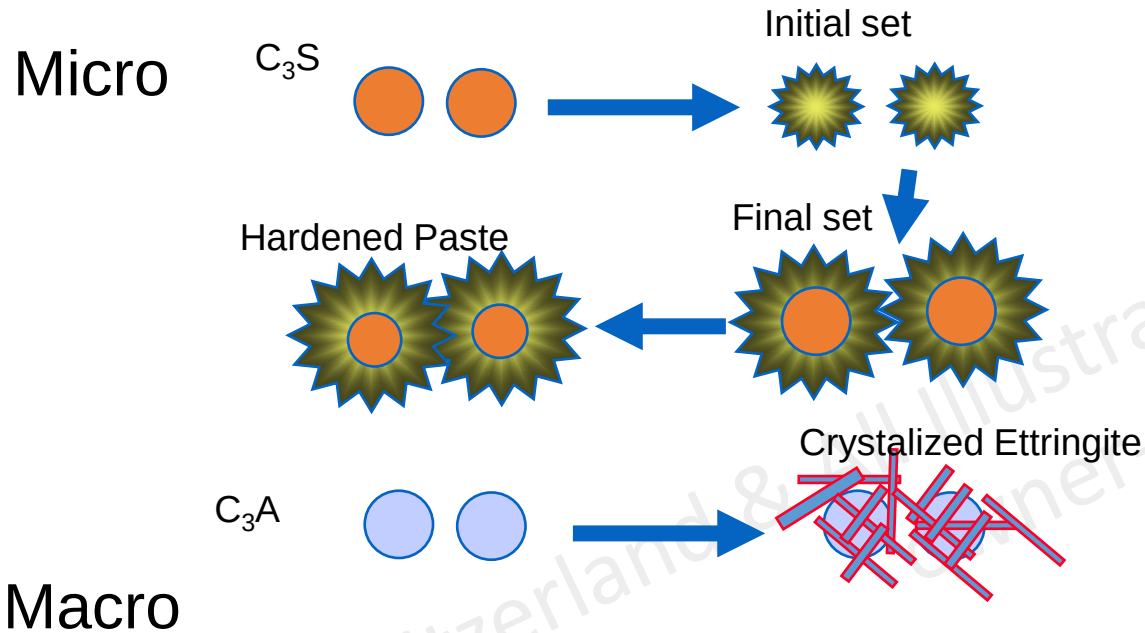
- Physical Effect

- *small particles*
- *particle packing*
- *micro filling*

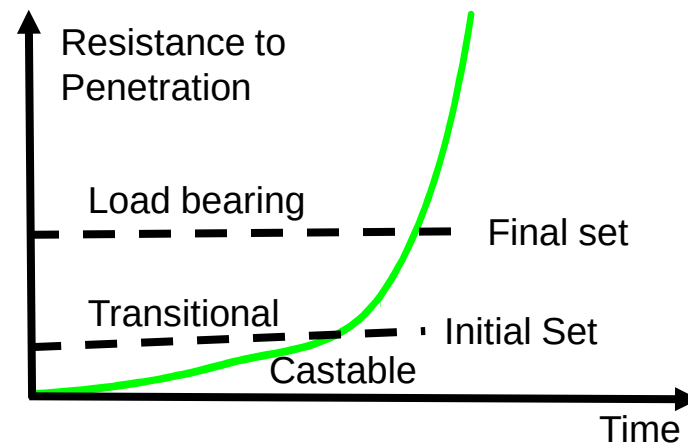
- Chemical Effect

- *pozzolanic material*
- *cement + water = CSH+ Calcium hydroxide*
- *pozzolan + calcium hydroxide + Water =CSH*
- *CSH= calcium silicate hydrate*

Mechanisms of Hydration

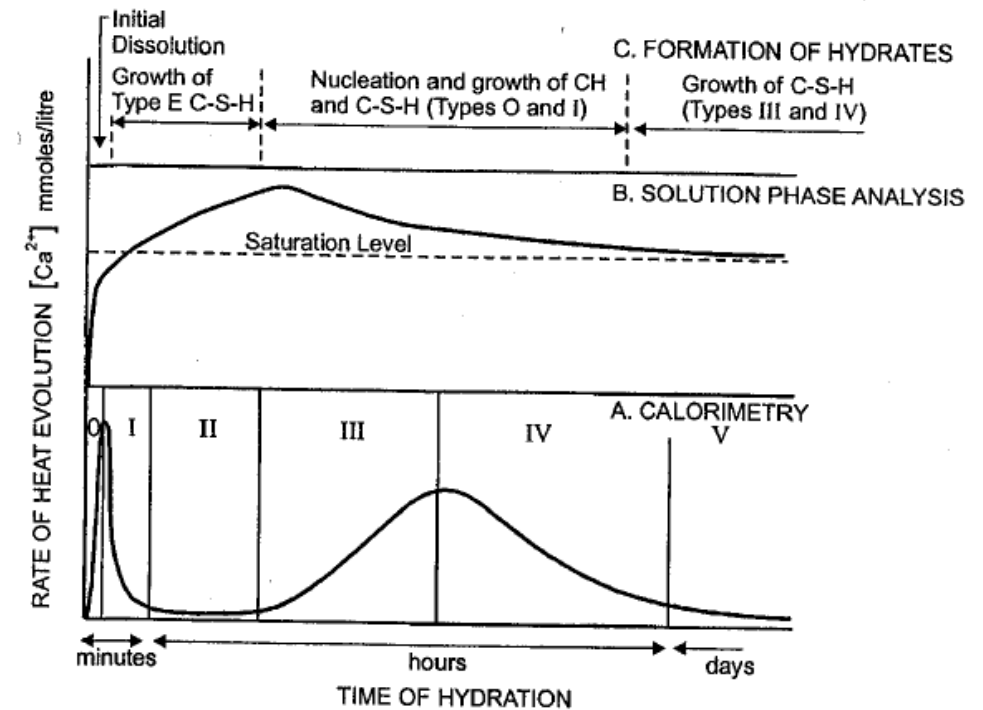


Macro



Low w/c ...

- improves density, strength
- decreases set time
- increases heat evolution
- decreases workability



Hydration and curing

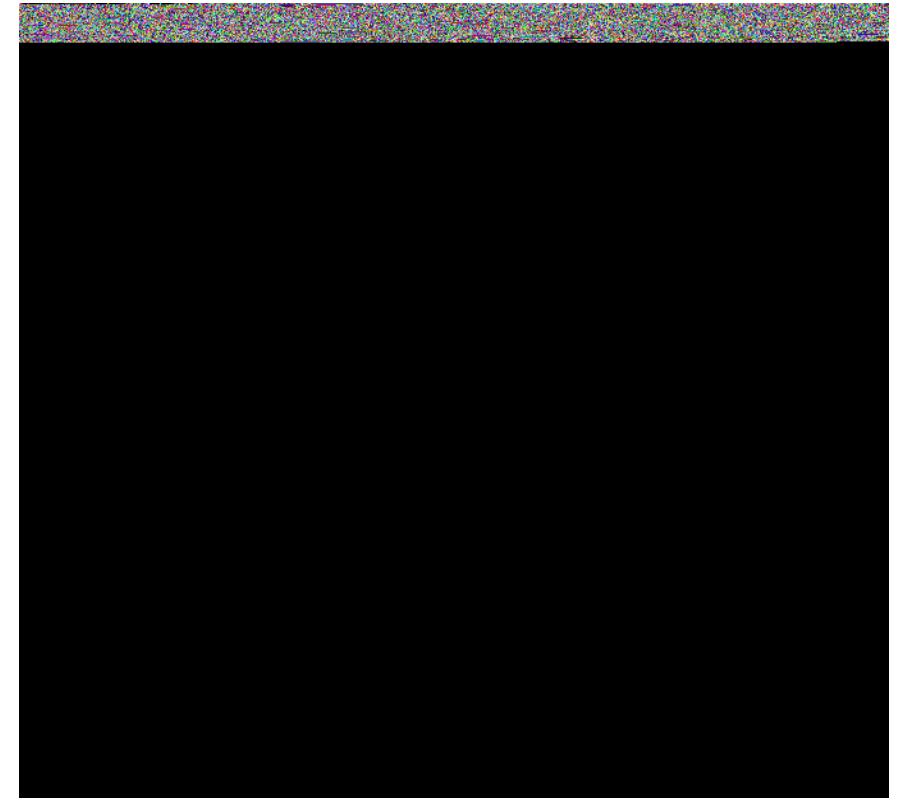
- Normal Curing, Accelerated Curing, Steam curing
- Maximum temperature of concrete curing, ACI specifications can be misused
 - Control parameters are not for the chamber, but differential concrete temperature, thermal cracking
 - Durability due to delayed ettringite formation
 - Can not start the curing until the set has taken place hydration has been initiated
- Motivated by High early strength due to accelerated curing and superplasticizers.
- Why will the long-term strength suffer from the accelerated curing (28-56 days strength)
- Silica fume, 2-4% air content 5-8%
- Plastic shrinkage cracking

Curing Concrete Objectives

1. Prevent loss of moisture (not less than 80% r.h.)
2. Maintain a favorable temperature for a definite period of time. (above freezing and below $176^{\circ}\text{F} = 80^{\circ}\text{C}$)
3. Time needed for the chemical reaction with water.
4. 14 days of curing, 40% of hydration is reached. 70 % at 28 days.

Effect of high temperatures on compressive strength at various ages (Klieger 1958).

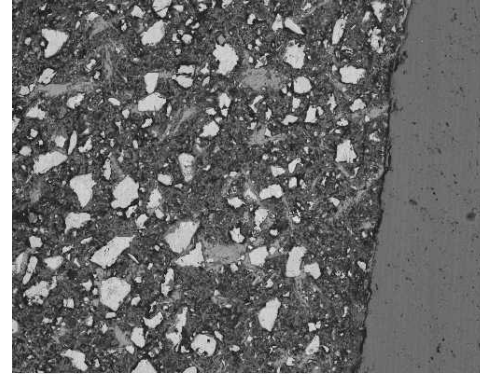
Effect of Concrete Temperatures on Strength



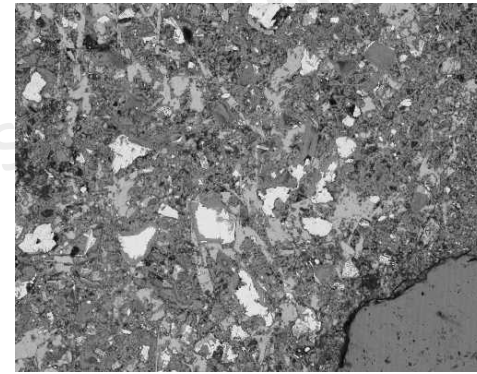
Interface Transition Zone, Calcium Hydroxide, and Ettringite

- Zone Between the aggregate surface and the hydrating cement microstructure.
- Weak zone, susceptible to cracking

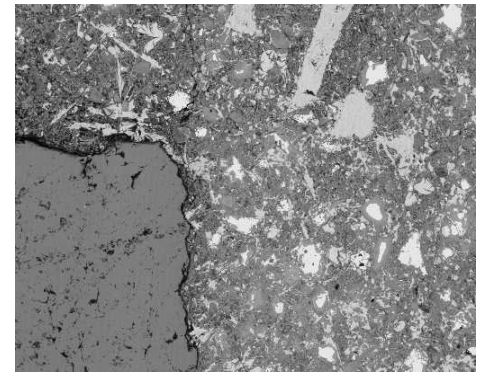
*Photos: Paul Stutzman, National
Institutes of Standards and Technology*



1 day



7 day



28 day

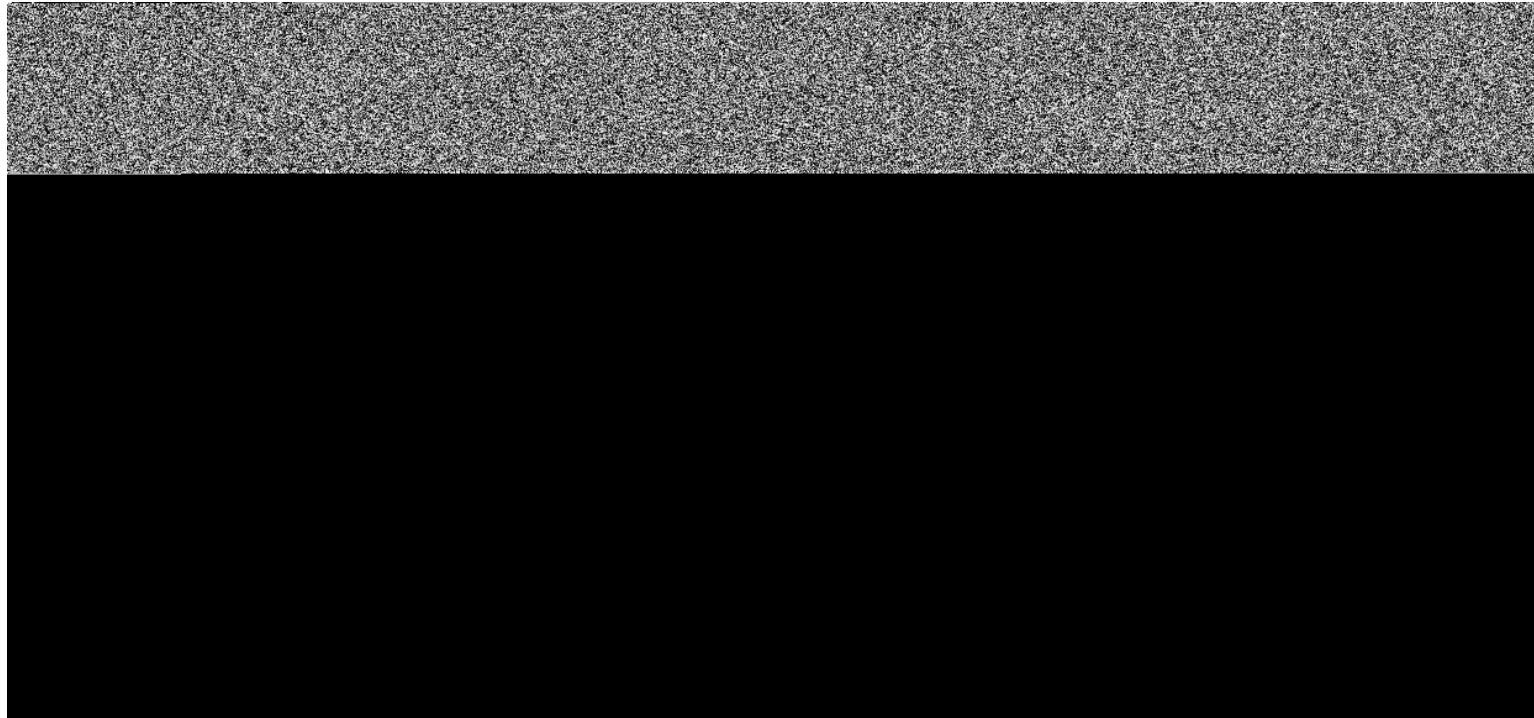
- Other Hydration Products



Effect of Silica Fume on the densification of the paste

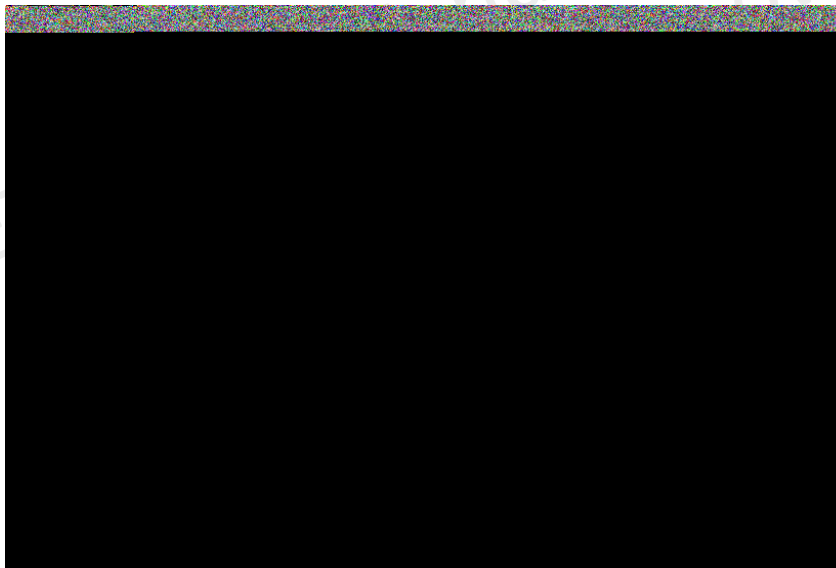
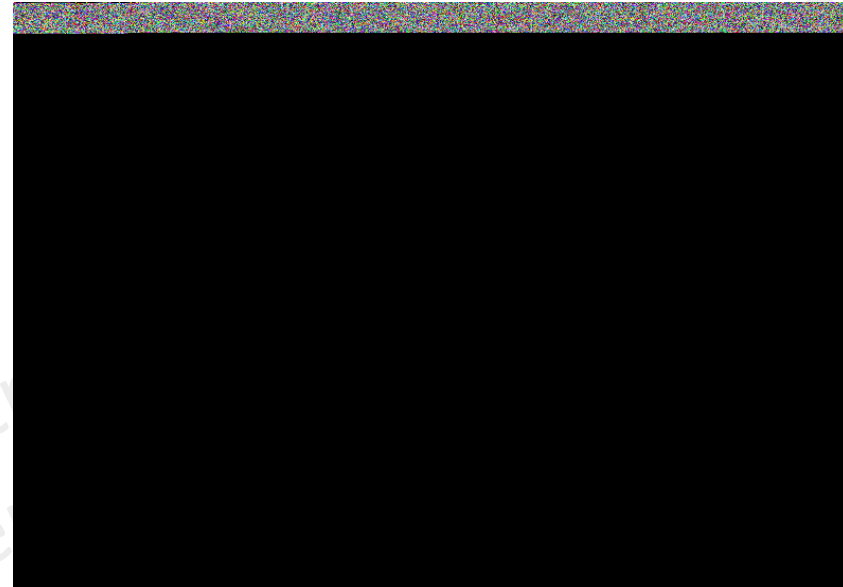
Paste

Paste+10% Silica Fume



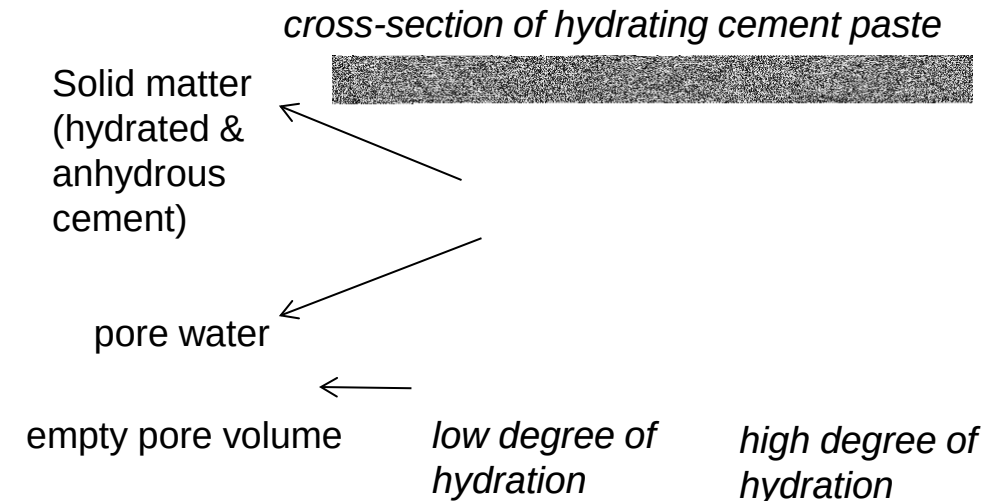
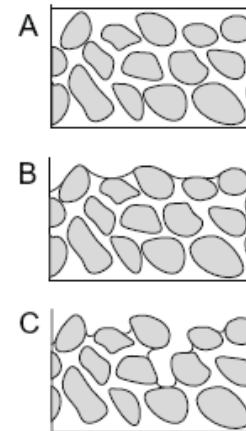
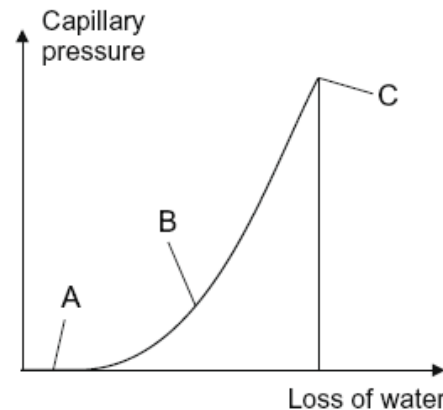
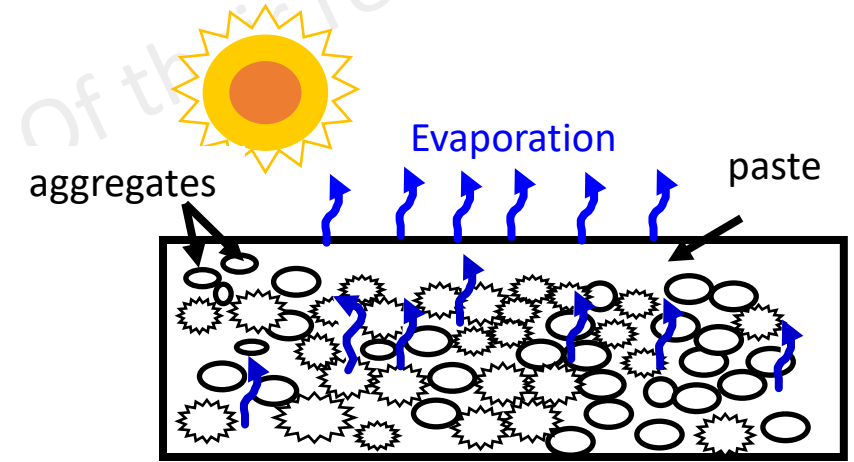
- 7 day old mortar (L0100 and L0103) showing (Field width of each is 80 microns, x1000)
- [David Lange, UIUC, and Hamlin Jennings and S.P. Shah, Northwestern University]

Interface Transition Zone

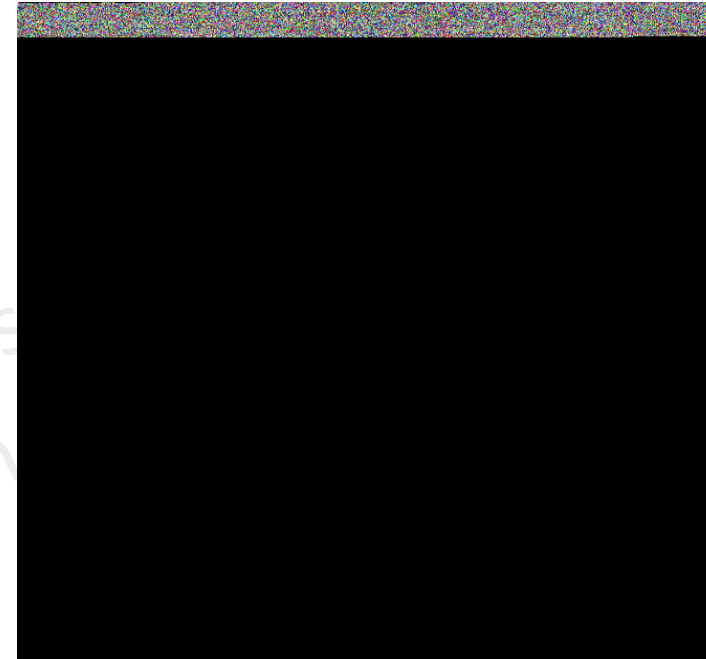
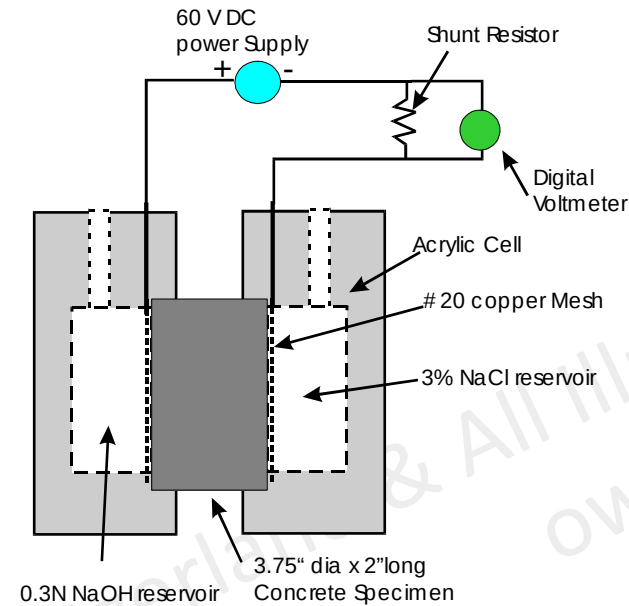


Shrinkage as a Volumetric Change in a restrained condition

- Drying and plastic shrinkage
(due to water loss by evaporation)
- Autogenous/self-desiccation shrinkage
(due to cement hydration)
 - Thermal shrinkage (due to decrease in temperature)
- Carbonation shrinkage
(reaction of hydrated cement w/carbon dioxide)

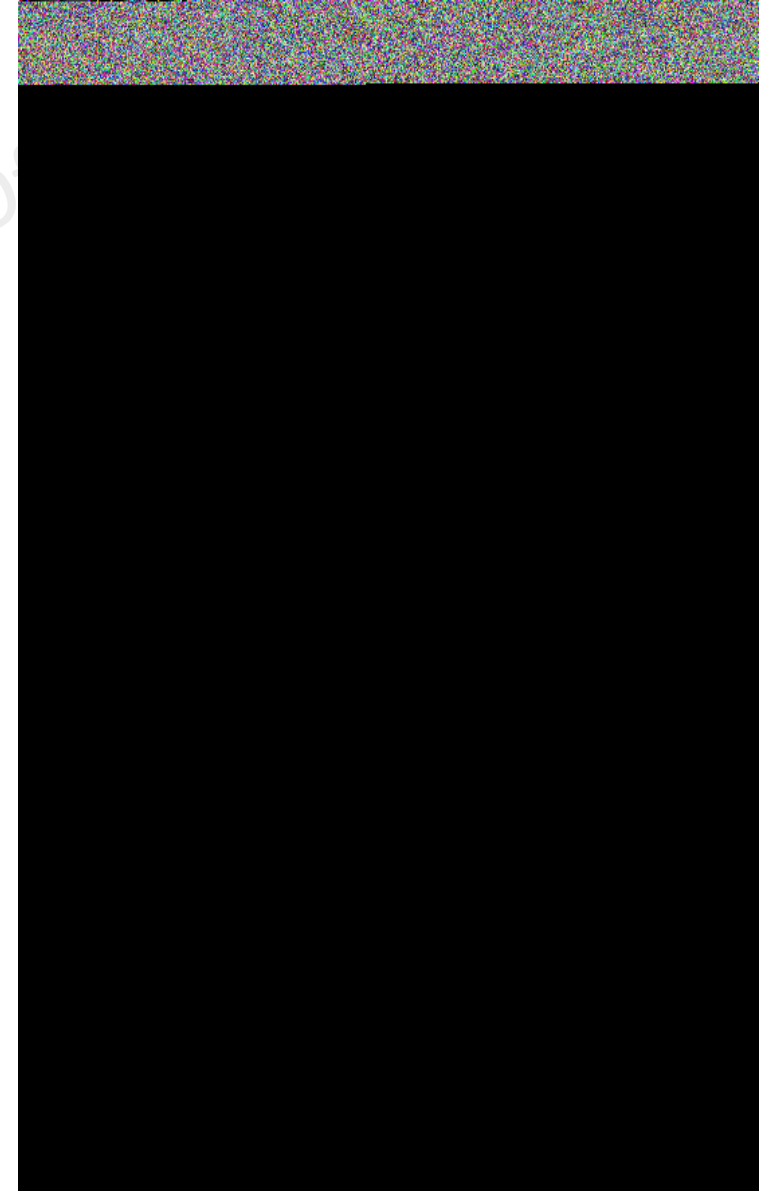
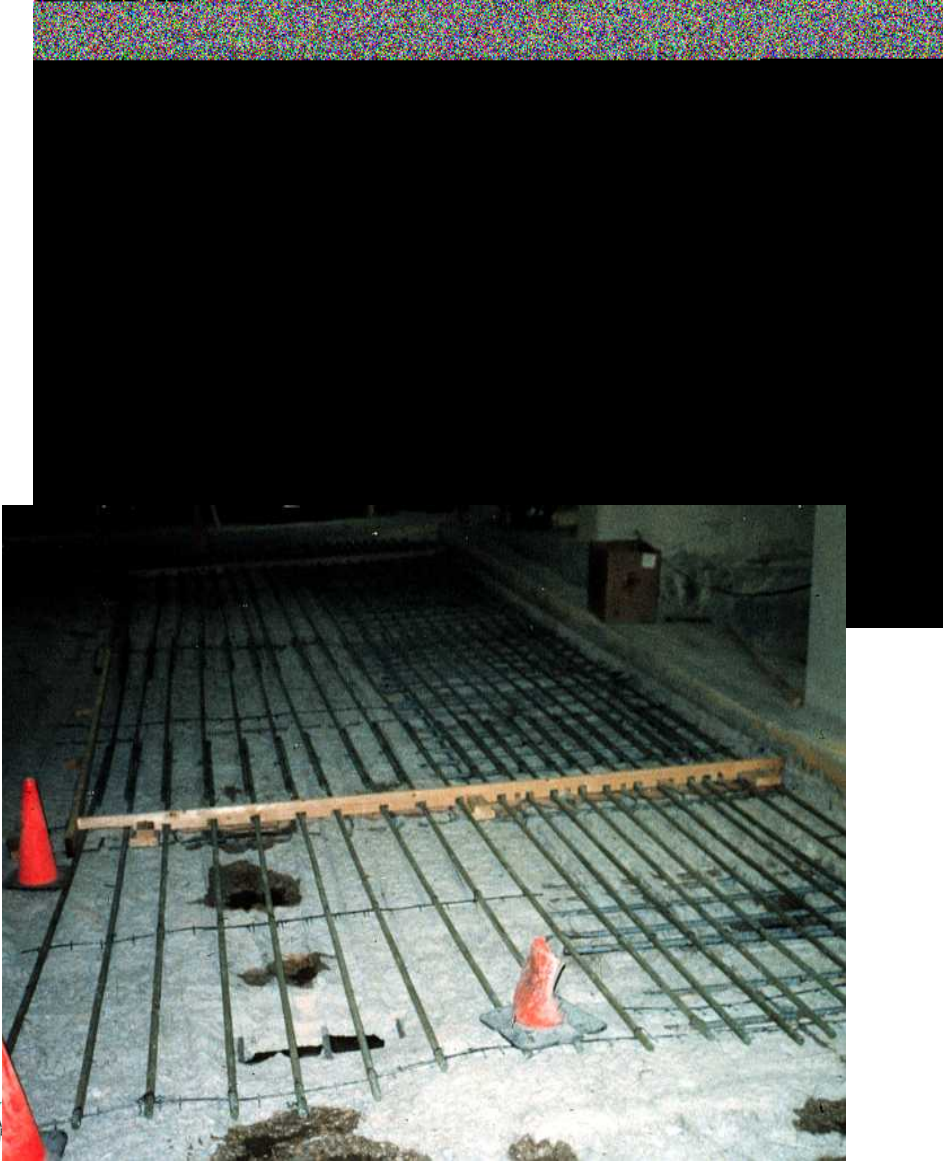


Rapid Chloride Permeability Test



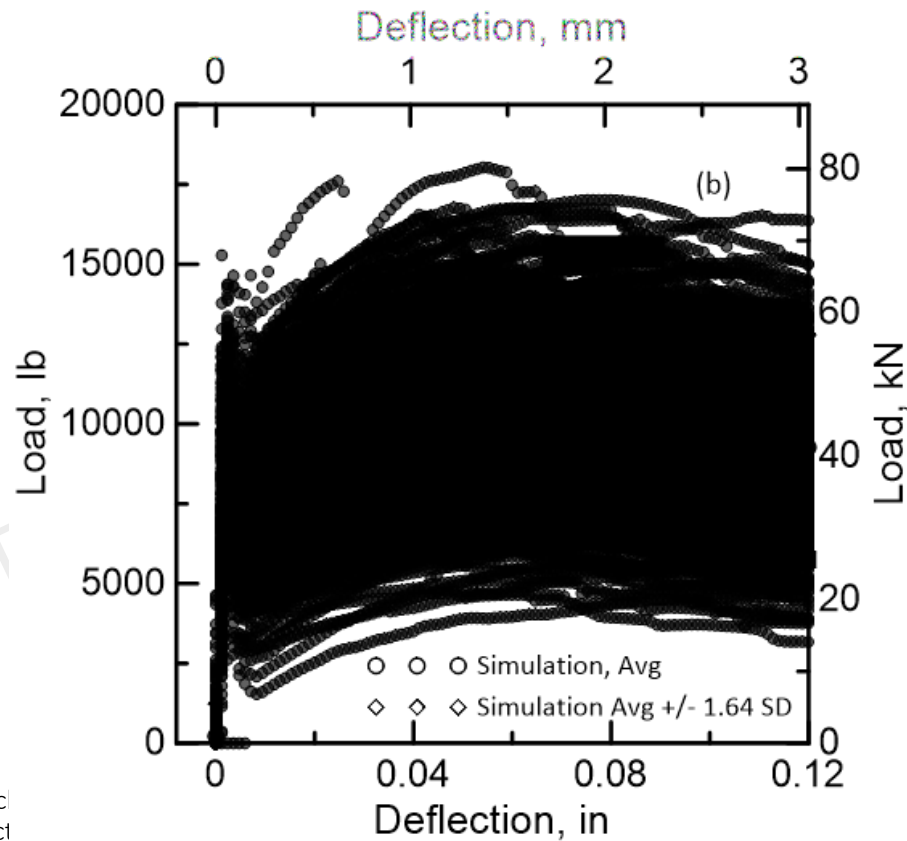
Silica fume	RCPT	Comp. Strength
0%	> 3,000 coulombs	= 5,000 psi
7-10%	< 1,000 coulombs	> 7,000 psi
>10%	< 500 coulombs	> 9,000 psi

Curing and Durability Aspects

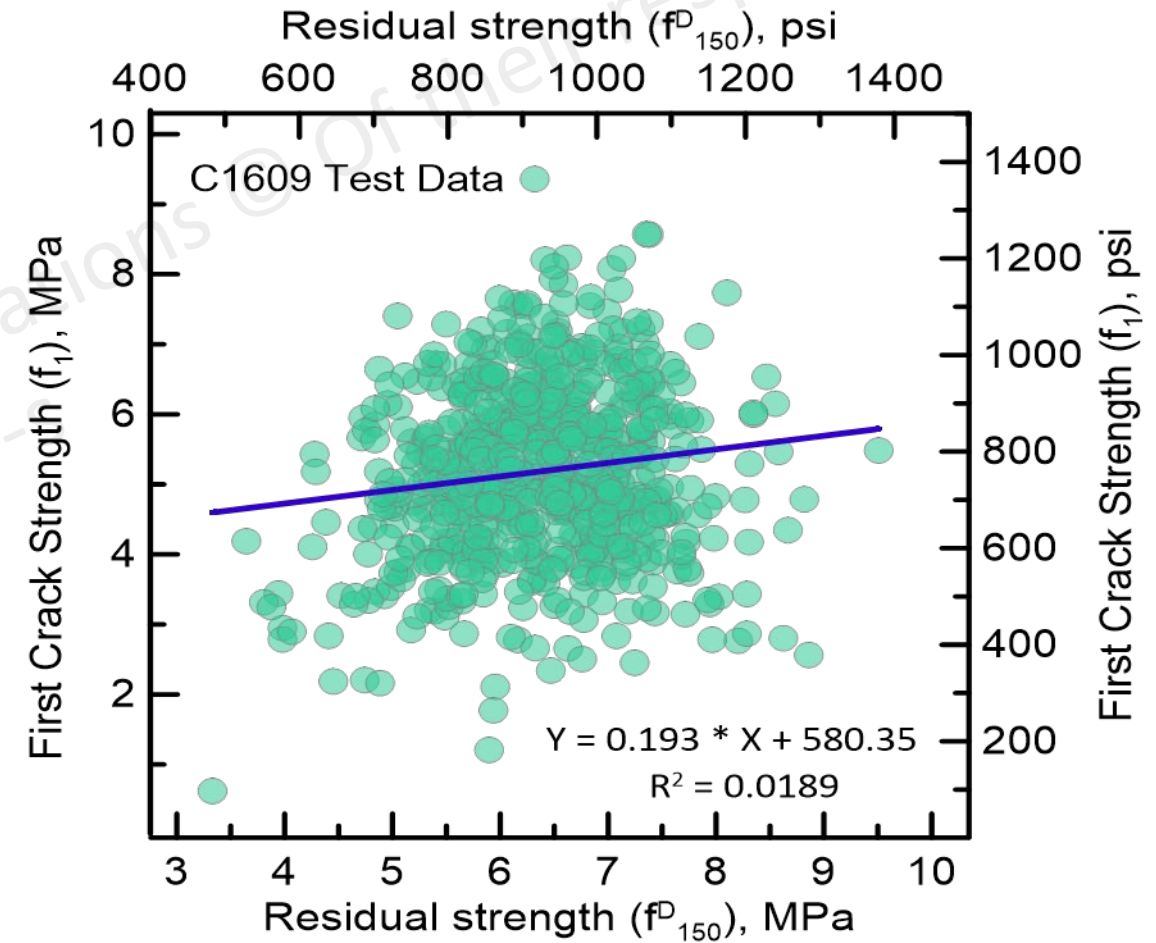
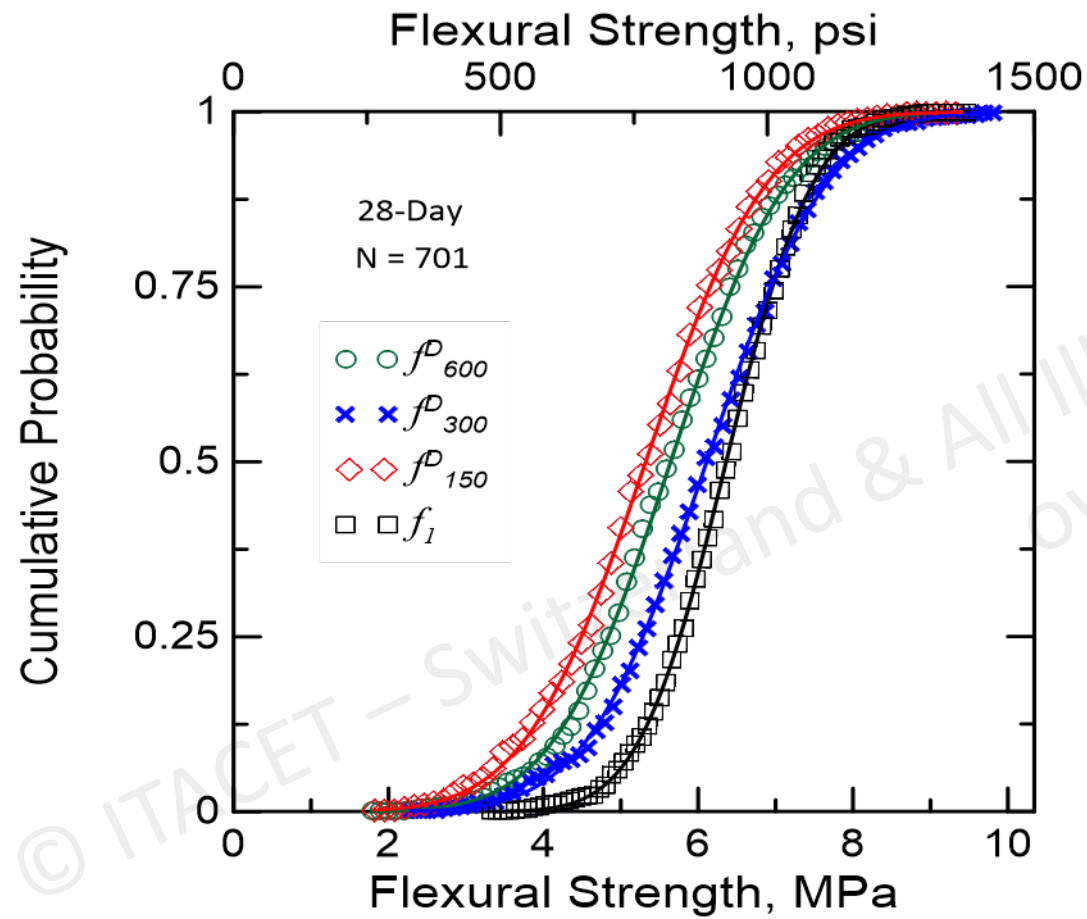


QC/QA Procedures for Precast Tunnel Linings

- Evaluation of more than 9 months of QC/QA beam samples, 1200+ flexural C1609 beams
- Address the variations using Statistical Process Control tools

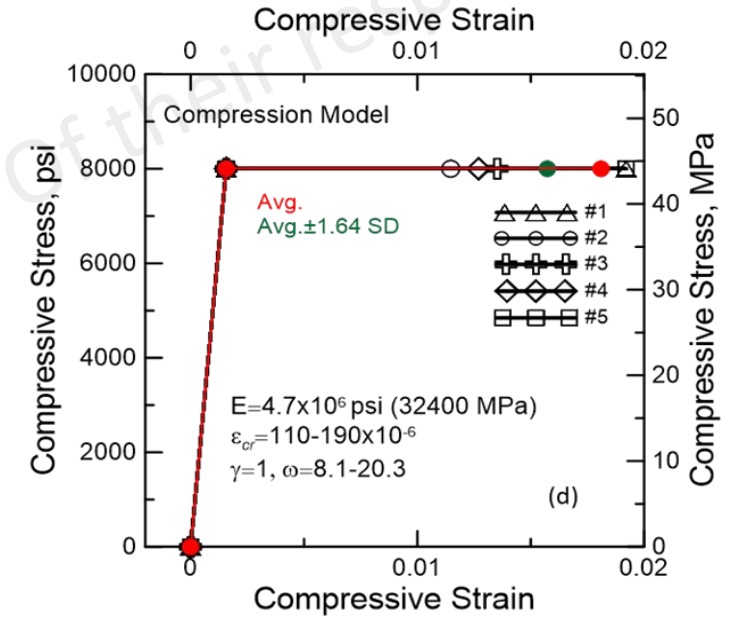
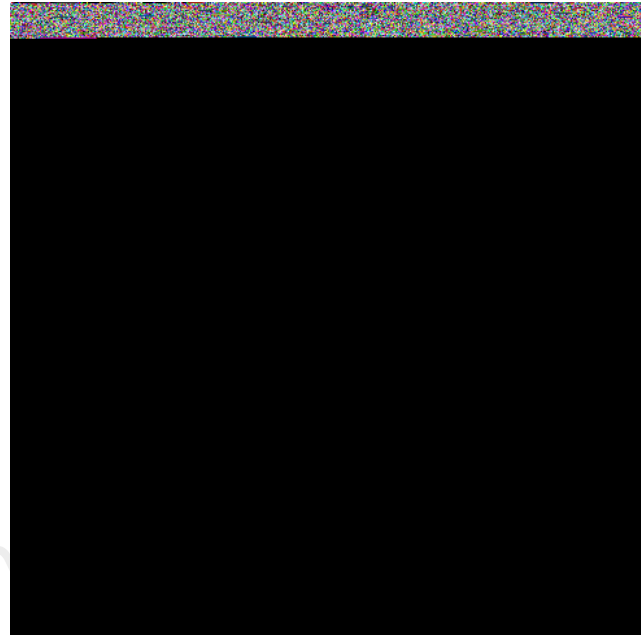
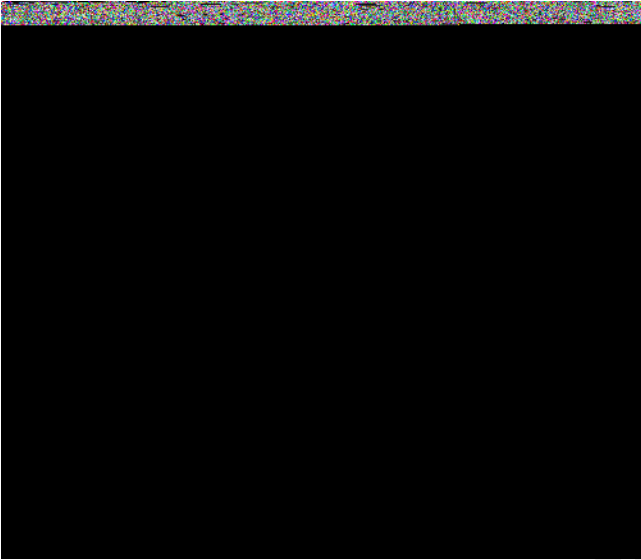


Statistical measures of distribution and correlation

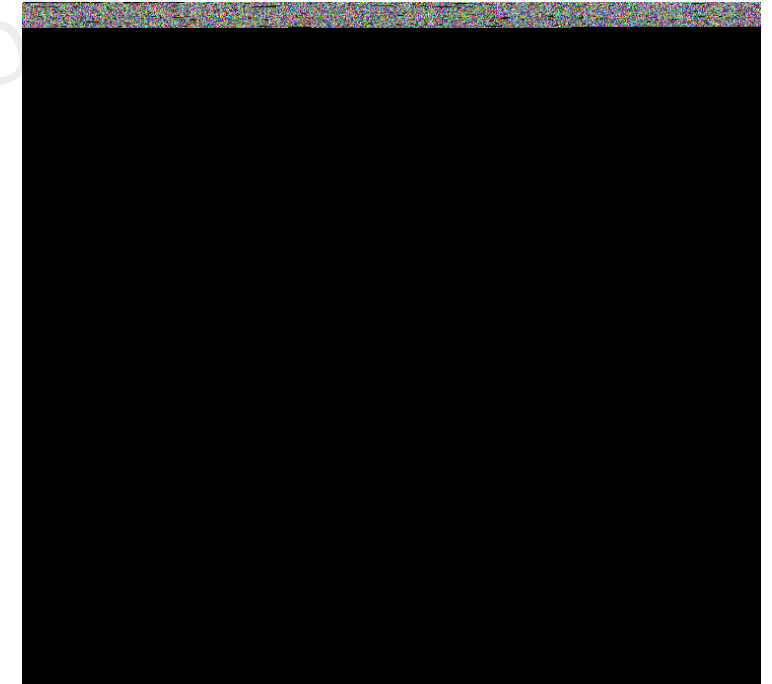
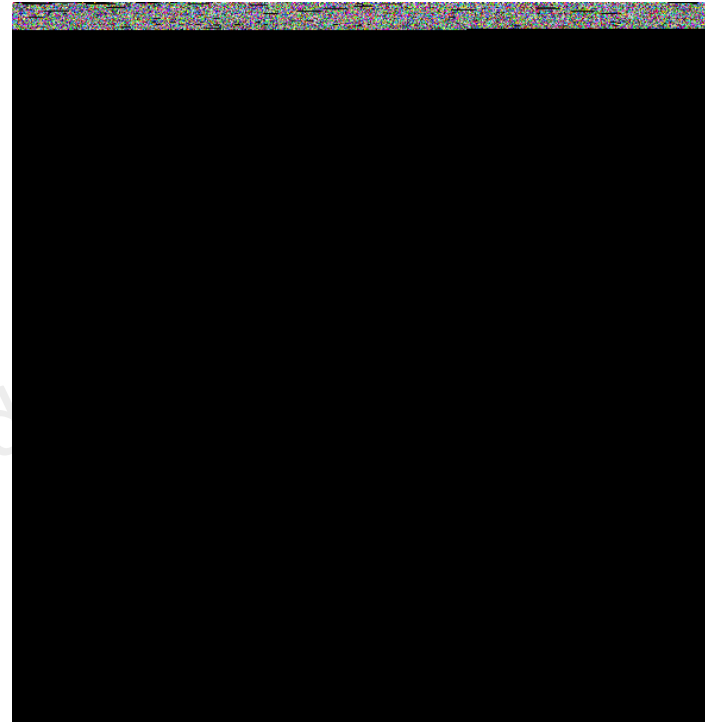
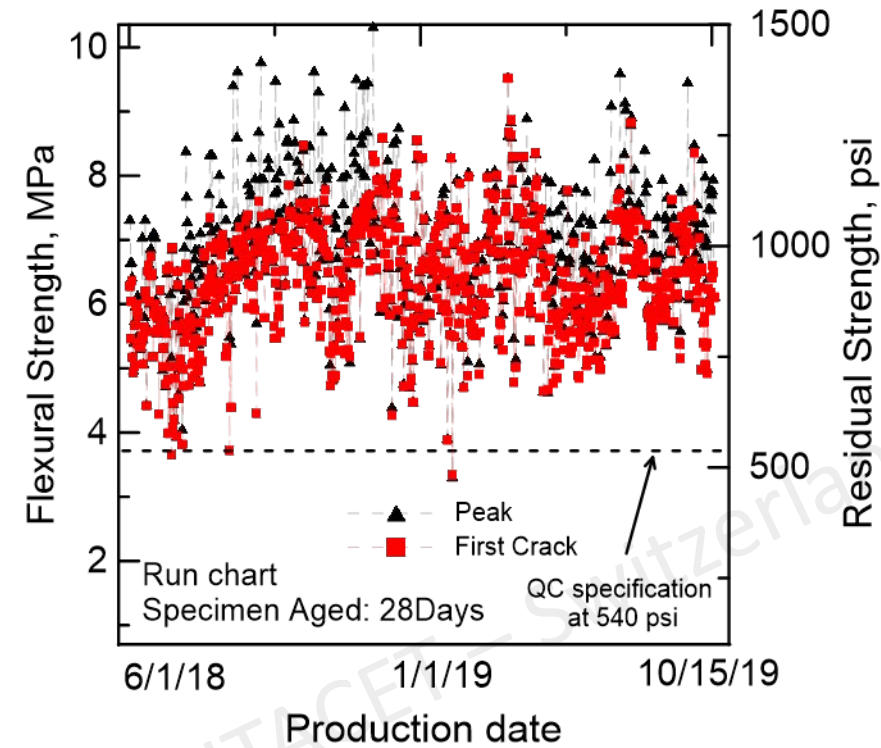


What do you do when there is no meaningful correlation?

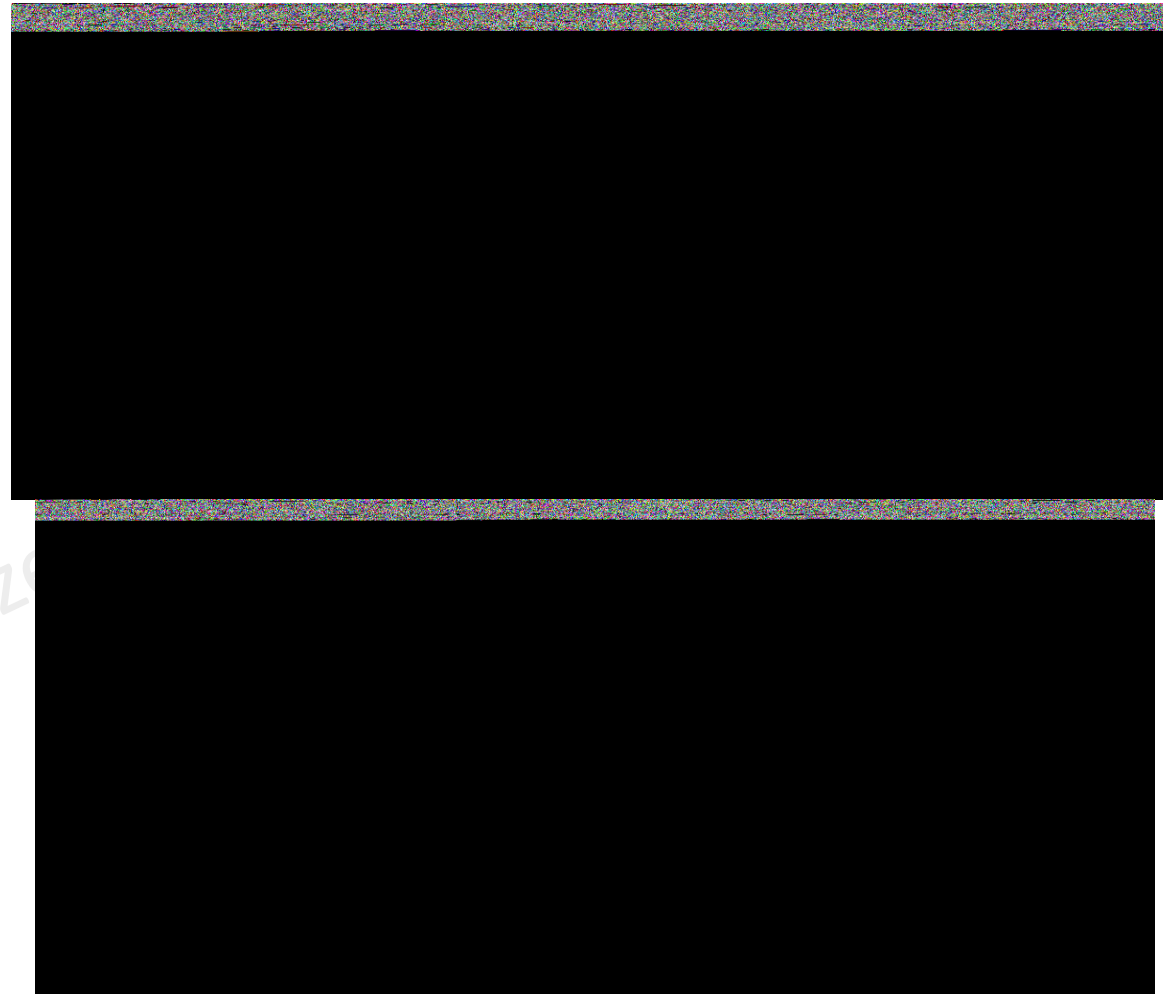
Simulation of 1200+ QC data in chart



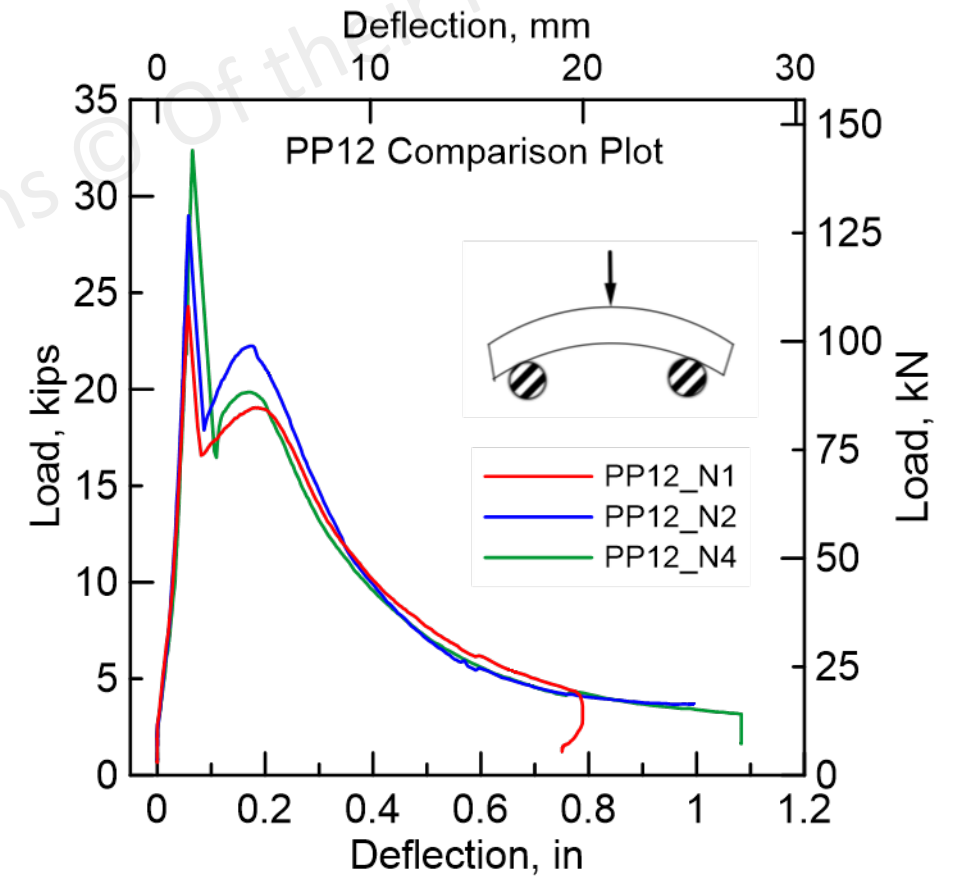
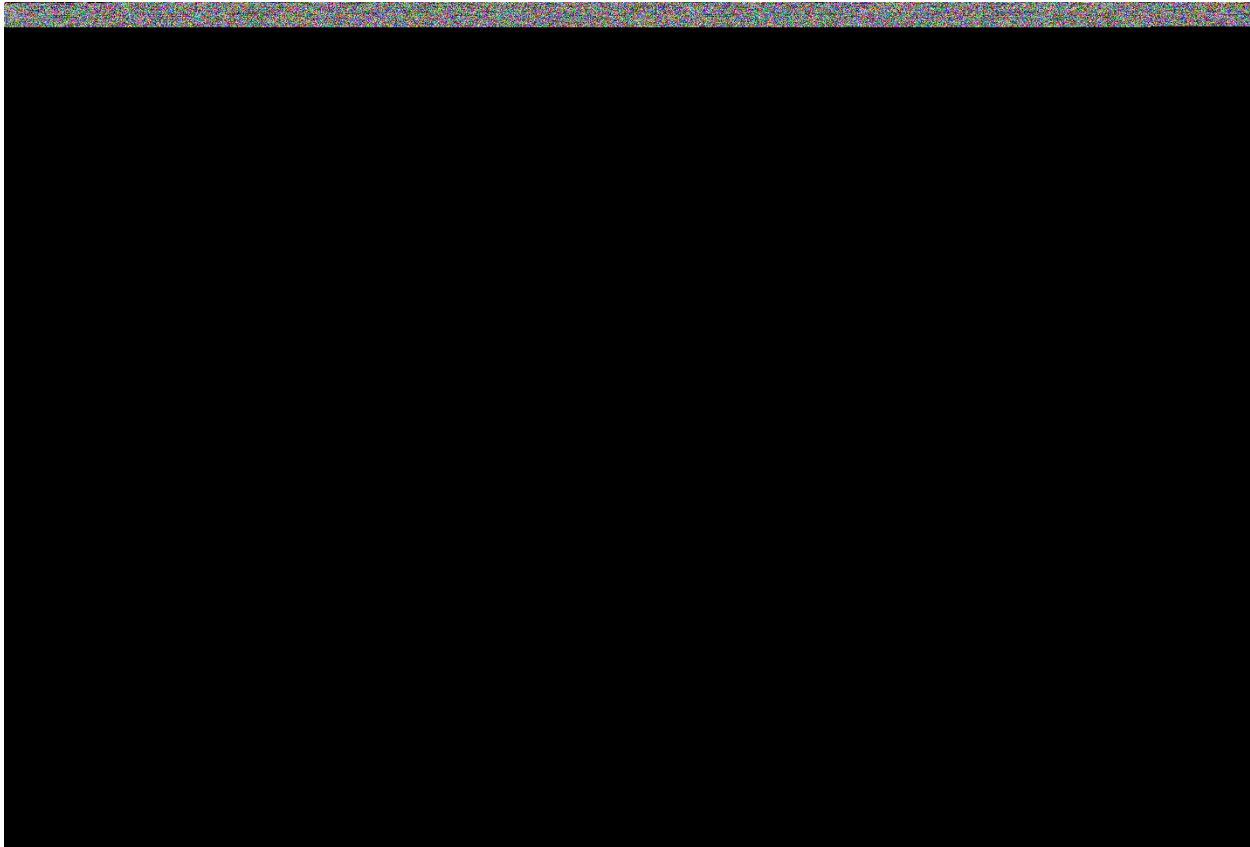
CUSUM and EWMA Run Charts to monitor and control Production problems and outlier analysis



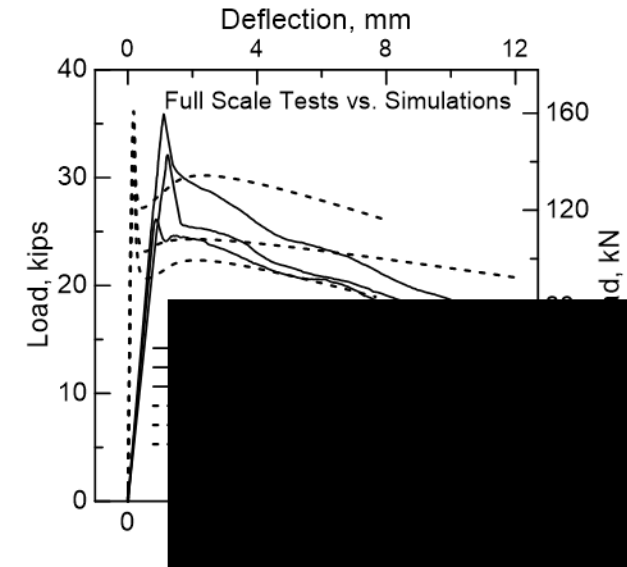
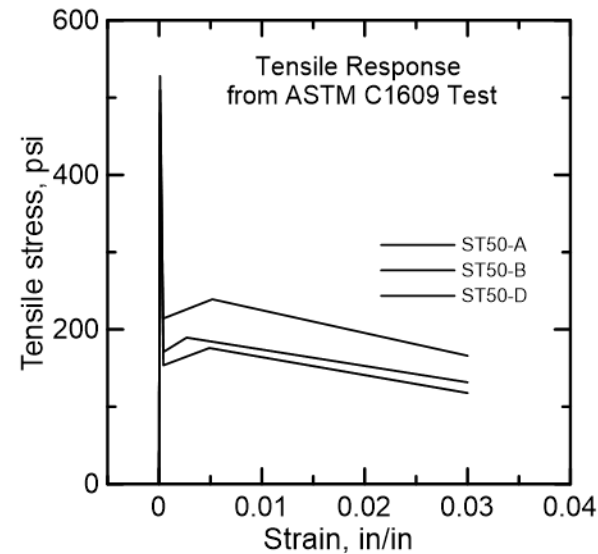
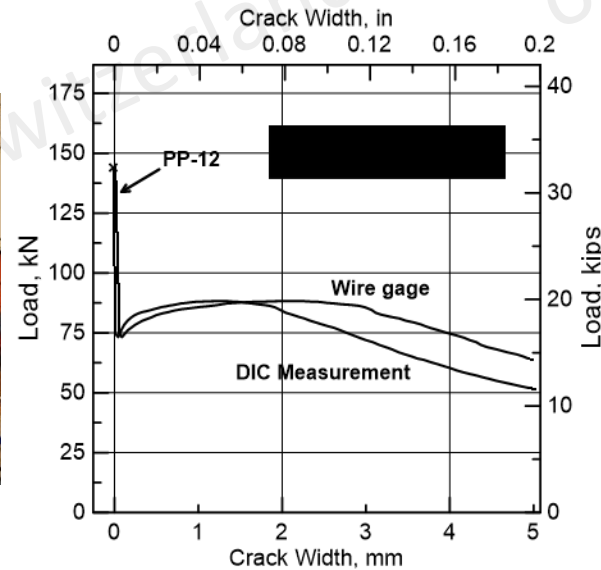
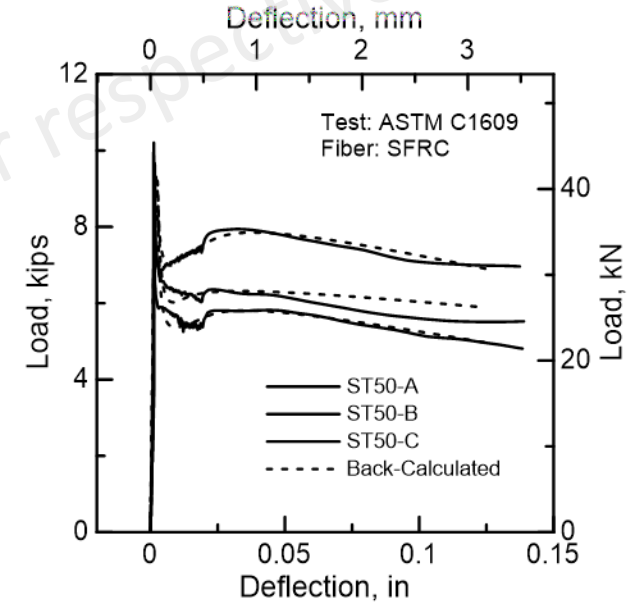
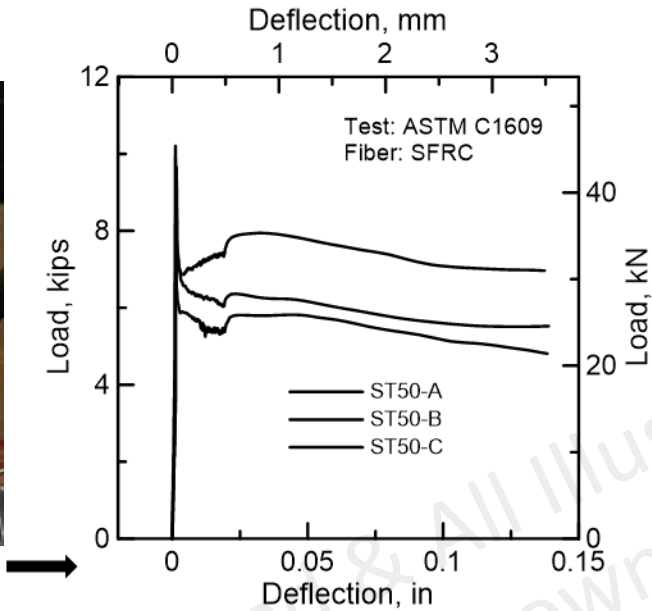
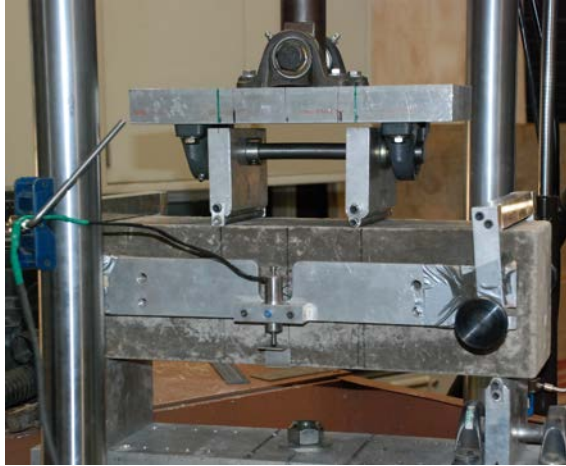
Control Charts for 9 months of production



Precast Tunnel Segment Testing Facilities at ASU



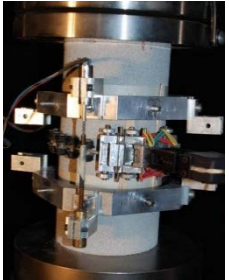
Quad-Linear Modeling for Simulating response of tunnel segments based on ASTM C1609 Tests



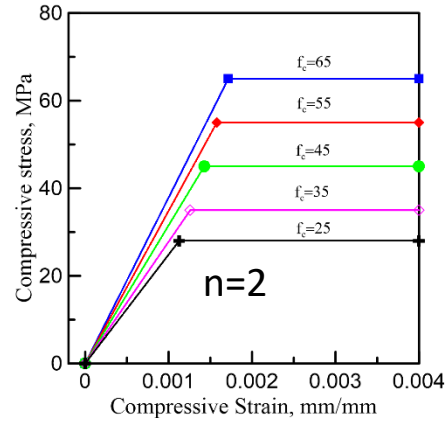
Mechanical Characterization and Design From laboratory to structure scale

Laboratory Scale

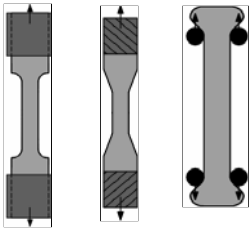
Compression



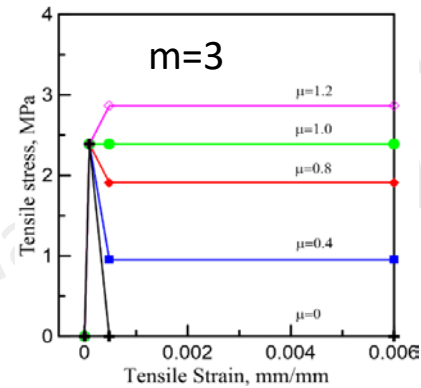
Uniaxial Compression



Tension

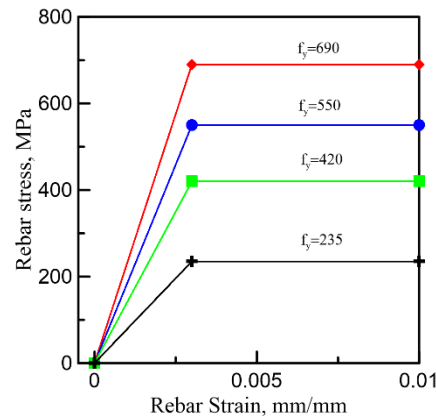


Uniaxial tension

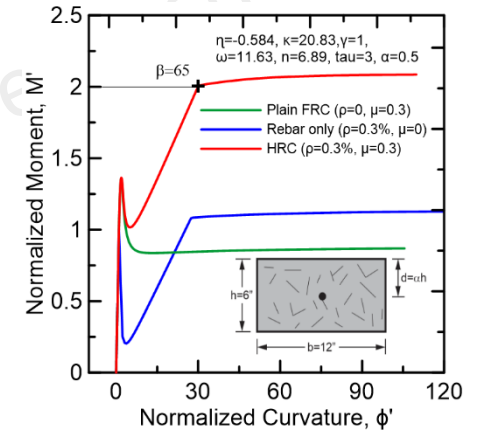
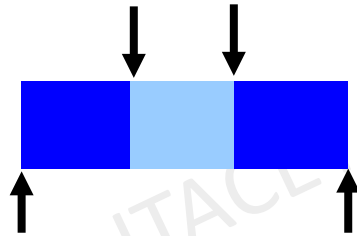


Flexural tests

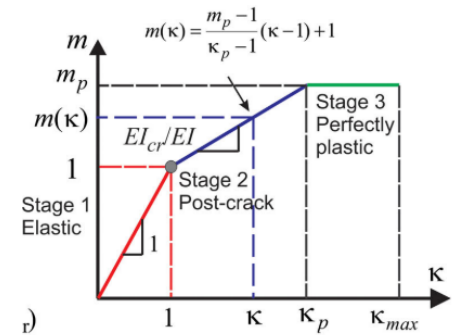
-ASTM C1609
-EN14561



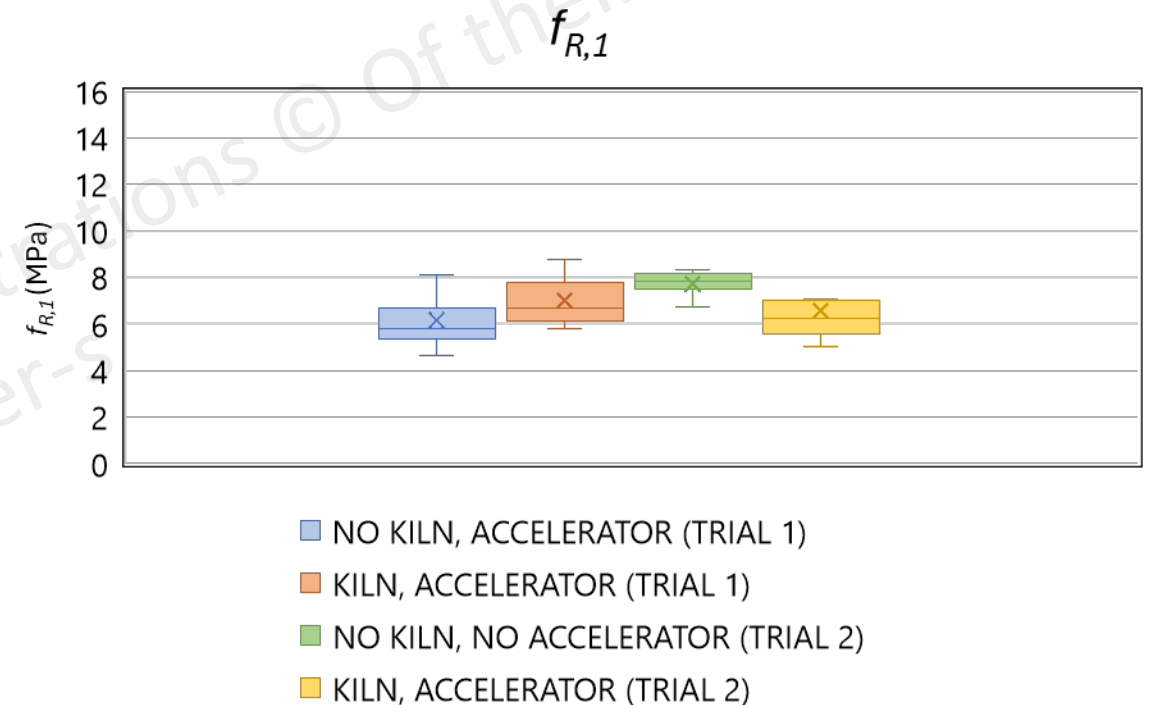
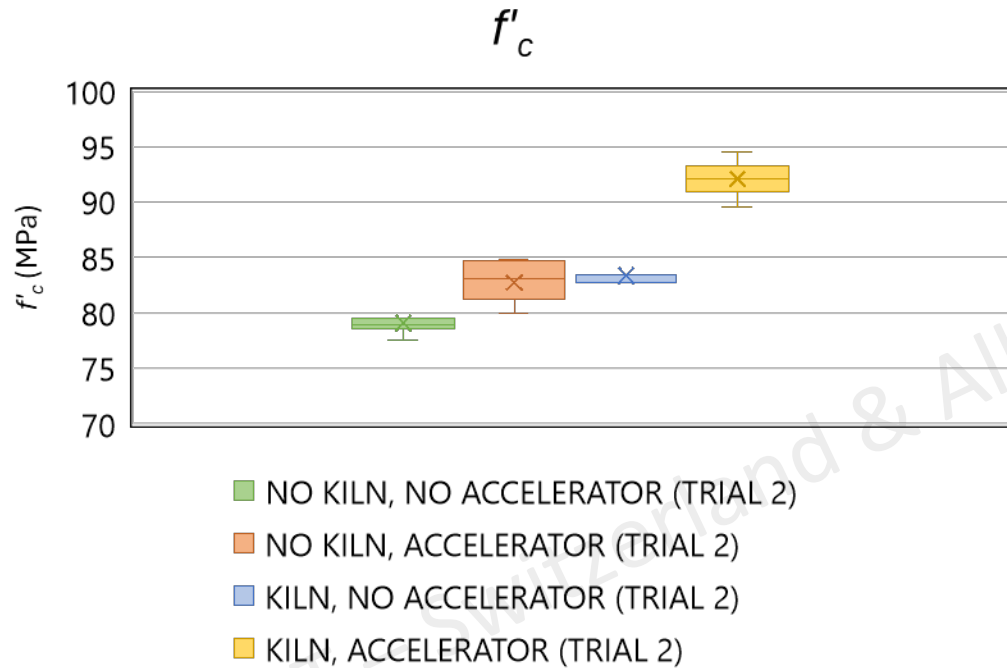
Reinforcement



Closed-Form Solutions

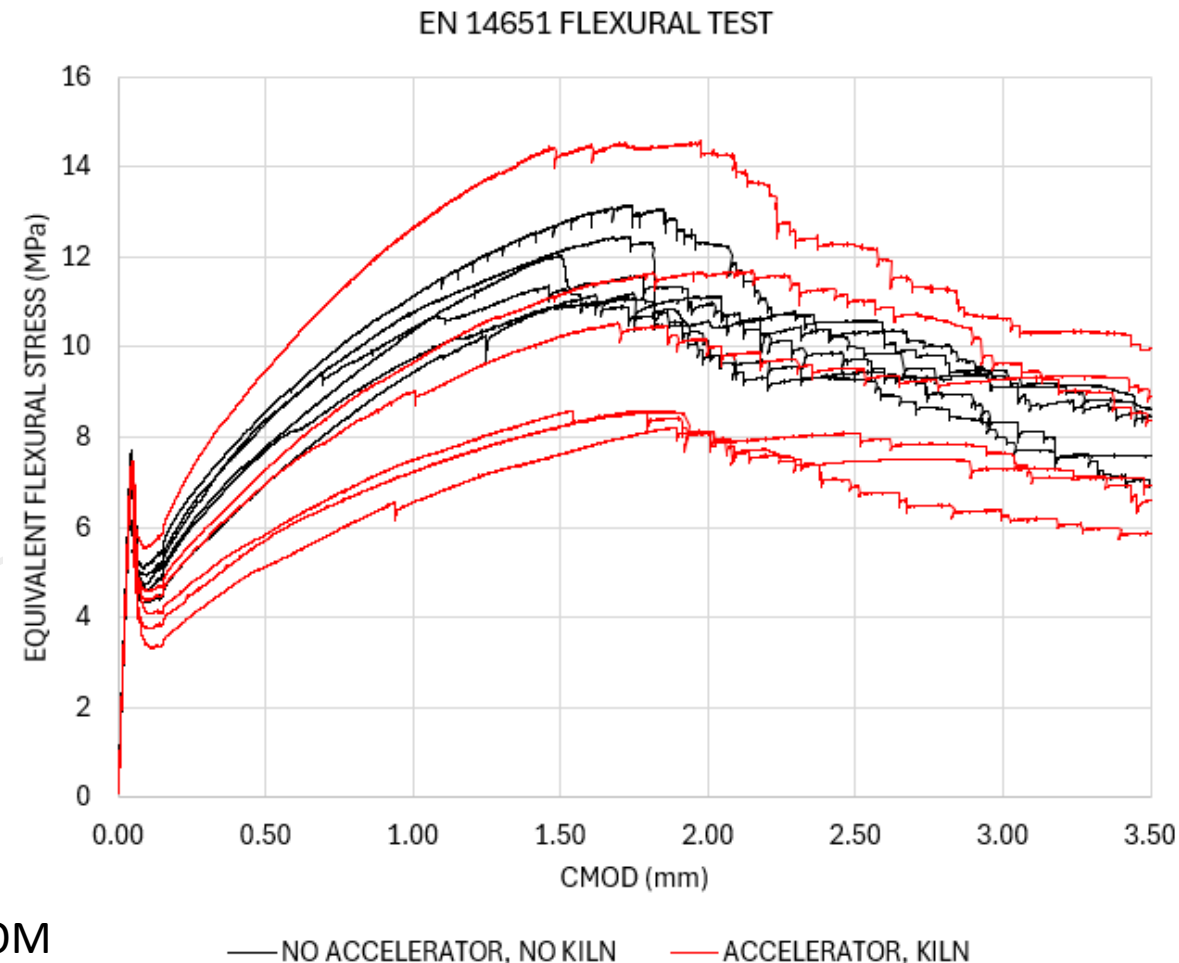


Effect of accelerator use and high temperature curing on the 28 day compressive strength Results



Data Courtesy of Dr. Bakhshi, AECOM

Effect of accelerator use and high temperature curing on Tunnel segments



Data Courtesy of Dr. Bakhshi, AECOM

Conclusions

- Properties of Fresh and Hardened Concrete are affected through the choice of cements used and the availability of supplementary cementitious products
- The hydration mechanisms and curing of concrete are affected by so many factors in the entire process, among them the cementitious components, aggregates, and mix design
- The Curing process in the early stages is affected by the temperature, Mineral, and Chemical Admixtures
- Field-based case studies can help with the QC and the sustainability aspects of concrete design